

# EE115B *Digital Circuits*

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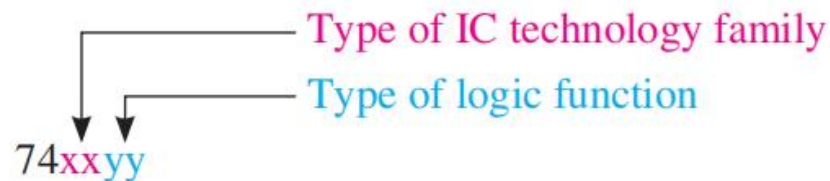


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# Fixed-Function Logic Gates

# 74 Series Logic Gate Functions

The 74 series is the standard fixed-function logic devices. The device label format includes one or more letters that indentify the type of logic circuit technology family in the IC package and two or more digits that identify the type of logic function. For example, 74HC04 is a fixed-function IC that has six inverters in a package as indicated by 04. The letters, HC, following the prefix 74 identify the circuit technology family as a type of CMOS logic.



HC: high speed CMOS

LS: Low power schottky (bipolar, TTL)

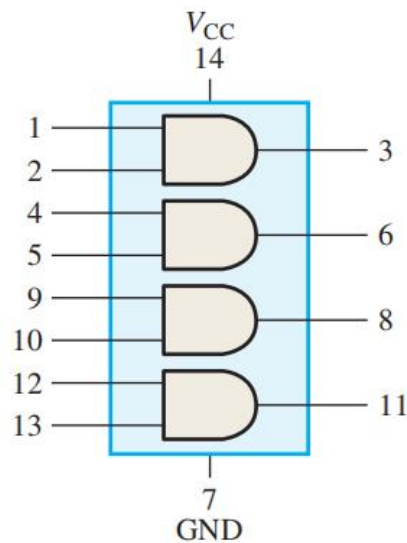
In CMOS vs bipolar (TTL) there are threshold differences that were incompatible.



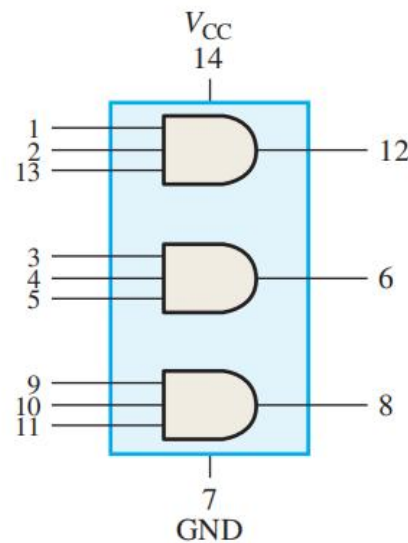
# Some Examples of Logic Gates

## AND Gate

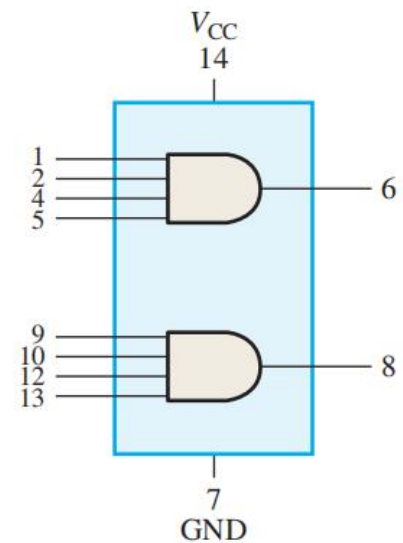
Figure 3–59 shows three configurations of fixed-function AND gates in the 74 series. The 74xx08 is a quad 2-input AND gate device, the 74xx11 is a triple 3-input AND gate device,



(a) 74xx08



(b) 74xx11



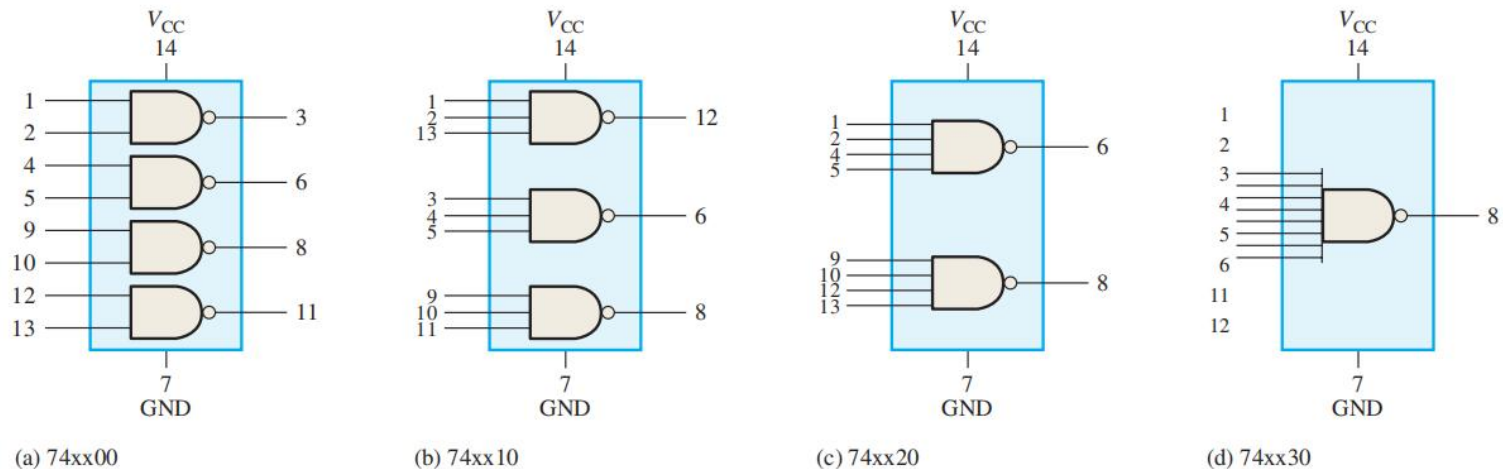
(c) 74xx21



# Some Examples of Logic Gates

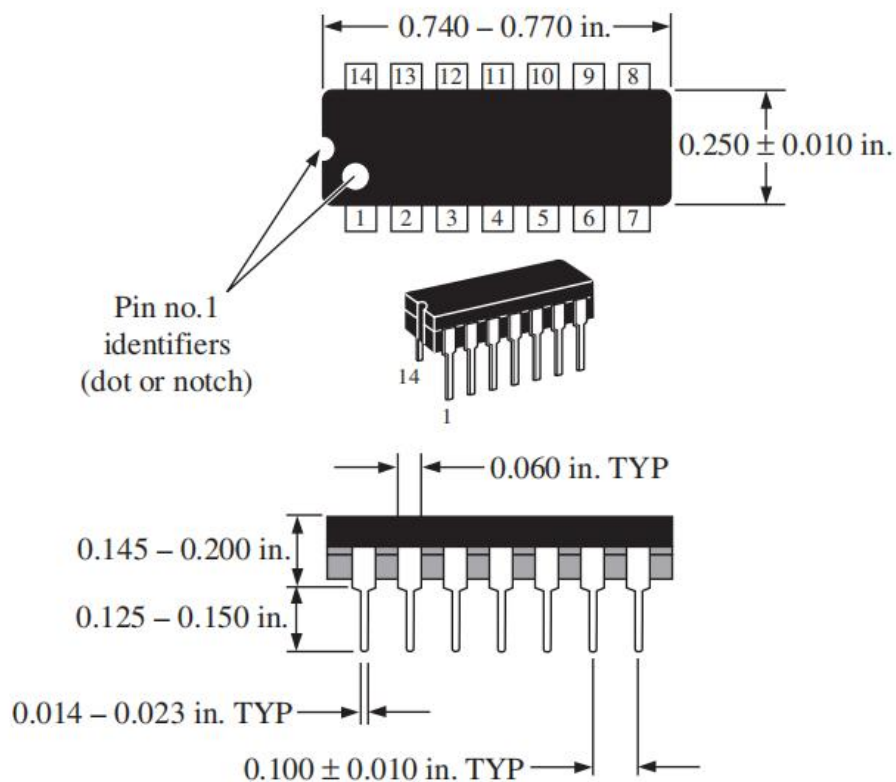
## NAND Gate

Figure 3–60 shows four configurations of fixed-function NAND gates in the 74 series. The 74xx00 is a quad 2-input NAND gate device, the 74xx10 is a triple 3-input NAND gate device, the 74xx20 is a dual 4-input NAND gate device, and the 74xx30 is a single 8-input NAND gate device.

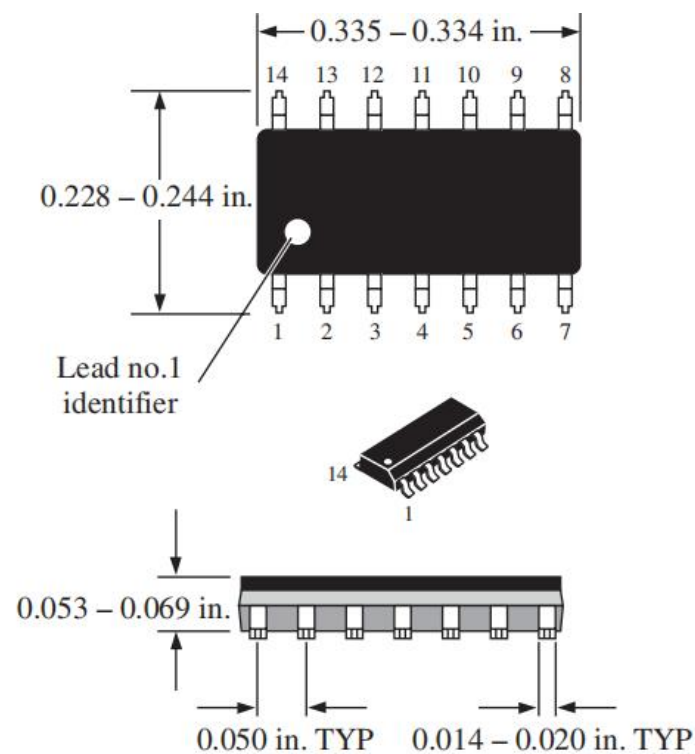


**FIGURE 3–60** 74 series NAND gate devices with package pin numbers.

# Packages



(a) 14-pin dual in-line package (DIP) for feedthrough mounting



(b) 14-pin small outline package (SOIC) for surface mounting



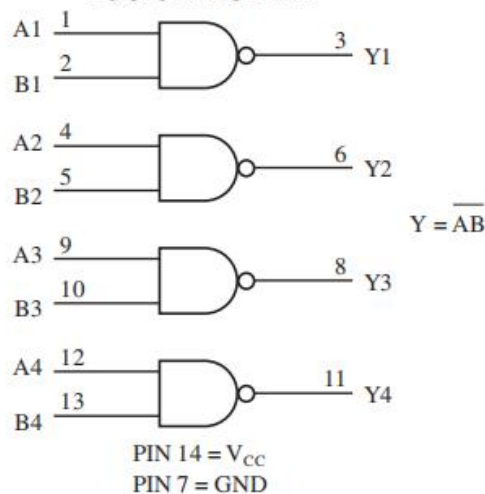
# Examples of Device Datasheet

## Quad 2-Input NAND Gate High-Performance Silicon-Gate CMOS

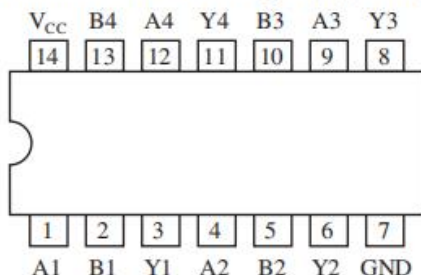
The MC54/74HC00A is identical in pinout to the LS00. The device inputs are compatible with Standard CMOS outputs; with pullup resistors, they are compatible with LSTTL outputs.

- Output Drive Capability: 10 LSTTL Loads
- Outputs Directly Interface to CMOS, NMOS and TTL
- Operating Voltage Range: 2 to 6 V
- Low Input Current:  $1\ \mu\text{A}$
- High Noise Immunity Characteristic of CMOS Devices
- In Compliance With the JEDEC Standard No. 7A Requirements
- Chip Complexity: 32 FETs or 8 Equivalent Gates

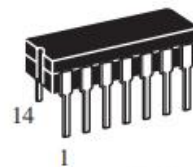
### LOGIC DIAGRAM



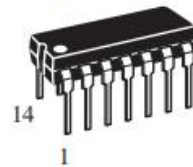
### Pinout: 14–Load Packages (Top View)



### MC54/74HC00A



J SUFFIX  
CERAMIC PACKAGE  
CASE 632-08



N SUFFIX  
PLASTIC PACKAGE  
CASE 646-06



D SUFFIX  
SOIC PACKAGE  
CASE 751A-03



DT SUFFIX  
TSSOP PACKAGE  
CASE 948G-01

### ORDERING INFORMATION

|             |         |
|-------------|---------|
| MC54HCXXAJ  | Ceramic |
| MC74HCXXAN  | Plastic |
| MC74HCXXAD  | SOIC    |
| MC74HCXXADT | TSSOP   |

### FUNCTION TABLE

| Inputs |   | Output |
|--------|---|--------|
| A      | B | Y      |
| L      | L | H      |
| L      | H | H      |
| H      | L | H      |
| H      | H | L      |



# Examples of Device Datasheet

## MAXIMUM RATINGS\*

| Symbol    | Parameter                                               | Value                  | Unit |
|-----------|---------------------------------------------------------|------------------------|------|
| $V_{CC}$  | DC Supply Voltage (Referenced to GND)                   | -0.5 to + 7.0          | V    |
| $V_{in}$  | DC Input Voltage (Referenced to GND)                    | -0.5 to $V_{CC} + 0.5$ | V    |
| $V_{out}$ | DC Output Voltage (Referenced to GND)                   | -0.5 to $V_{CC} + 0.5$ | V    |
| $I_{in}$  | DC Input Current, per Pin                               | $\pm 20$               | mA   |
| $I_{out}$ | DC Output Current, per Pin                              | $\pm 25$               | mA   |
| $I_{CC}$  | DC Supply Current, $V_{CC}$ and GND Pins                | $\pm 50$               | mA   |
| $P_D$     | Power Dissipation in Still Air, Plastic or Ceramic DIP† | 750                    | mW   |
|           | SOIC Package†                                           | 500                    |      |
|           | TSSOP Package†                                          | 450                    |      |
| $T_{stg}$ | Storage Temperature                                     | -65 to + 150           | °C   |
| $T_L$     | Lead Temperature, 1 mm from Case for 10 Seconds         |                        | °C   |
|           | Plastic DIP, SOIC or TSSOP Package                      | 260                    |      |
|           | Ceramic DIP                                             | 300                    |      |

\* Maximum Ratings are those values beyond which damage to the device may occur. Functional operation should be restricted to the Recommended Operating Conditions.

† Derating — Plastic DIP: - 10 mW/°C from 65° to 125° C  
 Ceramic DIP: - 10 mW/°C from 100° to 125° C  
 SOIC Package: - 7 mW/°C from 65° to 125° C  
 TSSOP Package: - 6.1 mW/°C from 65° to 125° C

# Examples of Device Datasheet

## RECOMMENDED OPERATING CONDITIONS

| Symbol            | Parameter                                            | in  | Max      | Unit |
|-------------------|------------------------------------------------------|-----|----------|------|
| $V_{CC}$          | DC Supply Voltage (Referenced to GND)                | 2.0 | 6.0      | V    |
| $V_{in}, V_{out}$ | DC Input Voltage, Output Voltage (Referenced to GND) | 0   | $V_{CC}$ | V    |
| $T_A$             | Operating Temperature, All Package Types             | -55 | +125     | °C   |
| $t_r, t_f$        | Input Rise and Fall Time                             |     |          |      |
|                   | $V_{CC} = 2.0\text{ V}$                              | 0   | 1000     | ns   |
|                   | $V_{CC} = 4.5\text{ V}$                              | 0   | 500      |      |
|                   | $V_{CC} = 6.0\text{ V}$                              | 0   | 400      |      |



# Examples of Device Datasheet

## DC CHARACTERISTICS (Voltages Referenced to GND)

MC54/74HC00A

| Symbol          | Parameter                                      | Condition                                                                                                                                         | V <sub>CC</sub><br>V | Guaranteed Limit |       |        | Unit |
|-----------------|------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------|----------------------|------------------|-------|--------|------|
|                 |                                                |                                                                                                                                                   |                      | -55 to 25°C      | ≤85°C | ≤125°C |      |
| V <sub>IH</sub> | Minimum High-Level Input Voltage               | V <sub>out</sub> = 0.1V or V <sub>CC</sub> - 0.1V<br> I <sub>out</sub>   ≤ 20μA                                                                   | 2.0                  | 1.50             | 1.50  | 1.50   | V    |
|                 |                                                |                                                                                                                                                   | 3.0                  | 2.10             | 2.10  | 2.10   |      |
|                 |                                                |                                                                                                                                                   | 4.5                  | 3.15             | 3.15  | 3.15   |      |
|                 |                                                |                                                                                                                                                   | 6.0                  | 4.20             | 4.20  | 4.20   |      |
| V <sub>IL</sub> | Maximum Low-Level Input Voltage                | V <sub>out</sub> = 0.1V or V <sub>CC</sub> - 0.1V<br> I <sub>out</sub>   ≤ 20μA                                                                   | 2.0                  | 0.50             | 0.50  | 0.50   | V    |
|                 |                                                |                                                                                                                                                   | 3.0                  | 0.90             | 0.90  | 0.90   |      |
|                 |                                                |                                                                                                                                                   | 4.5                  | 1.35             | 1.35  | 1.35   |      |
|                 |                                                |                                                                                                                                                   | 6.0                  | 1.80             | 1.80  | 1.80   |      |
| V <sub>OH</sub> | Minimum High-Level Output Voltage              | V <sub>in</sub> = V <sub>IH</sub> or V <sub>IL</sub><br> I <sub>out</sub>   ≤ 20μA                                                                | 2.0                  | 1.9              | 1.9   | 1.9    | V    |
|                 |                                                |                                                                                                                                                   | 4.5                  | 4.4              | 4.4   | 4.4    |      |
|                 |                                                |                                                                                                                                                   | 6.0                  | 5.9              | 5.9   | 5.9    |      |
|                 |                                                | V <sub>in</sub> = V <sub>IH</sub> or V <sub>IL</sub><br> I <sub>out</sub>   ≤ 2.4mA<br> I <sub>out</sub>   ≤ 4.0mA<br> I <sub>out</sub>   ≤ 5.2mA | 3.0                  | 2.48             | 2.34  | 2.20   |      |
|                 |                                                |                                                                                                                                                   | 4.5                  | 3.98             | 3.84  | 3.70   |      |
|                 |                                                |                                                                                                                                                   | 6.0                  | 5.48             | 5.34  | 5.20   |      |
| V <sub>OL</sub> | Maximum Low-Level Output Voltage               | V <sub>in</sub> = V <sub>IH</sub> or V <sub>IL</sub><br> I <sub>out</sub>   ≤ 20μA                                                                | 2.0                  | 0.1              | 0.1   | 0.1    | V    |
|                 |                                                |                                                                                                                                                   | 4.5                  | 0.1              | 0.1   | 0.1    |      |
|                 |                                                |                                                                                                                                                   | 6.0                  | 0.1              | 0.1   | 0.1    |      |
|                 |                                                | V <sub>in</sub> = V <sub>IH</sub> or V <sub>IL</sub><br> I <sub>out</sub>   ≤ 2.4mA<br> I <sub>out</sub>   ≤ 4.0mA<br> I <sub>out</sub>   ≤ 5.2mA | 3.0                  | 0.26             | 0.33  | 0.40   |      |
|                 |                                                |                                                                                                                                                   | 4.5                  | 0.26             | 0.33  | 0.40   |      |
|                 |                                                |                                                                                                                                                   | 6.0                  | 0.26             | 0.33  | 0.40   |      |
| I <sub>in</sub> | Maximum Input Leakage Current                  | V <sub>in</sub> = V <sub>CC</sub> or GND                                                                                                          | 6.0                  | ±0.1             | ±1.0  | ±1.0   | μA   |
| I <sub>CC</sub> | Maximum Quiescent Supply Current (per Package) | V <sub>in</sub> = V <sub>CC</sub> or GND<br>I <sub>out</sub> = 0μA                                                                                | 6.0                  | 1.0              | 10    | 40     | μA   |



# Examples of Device Datasheet

AC CHARACTERISTICS ( $C_L = 50 \text{ pF}$ , Input  $t_r = t_f = 6 \text{ ns}$ )

| Symbol                   | Parameter                                           | $V_{CC}$<br>V | Guaranteed Limit |       |        | Unit |
|--------------------------|-----------------------------------------------------|---------------|------------------|-------|--------|------|
|                          |                                                     |               | -55 to 25°C      | ≤85°C | ≤125°C |      |
| $t_{PLH}$ ,<br>$t_{PHL}$ | Maximum Propagation Delay, Input A or B to Output Y | 2.0           | 75               | 95    | 110    | ns   |
|                          |                                                     | 3.0           | 30               | 40    | 55     |      |
|                          |                                                     | 4.5           | 15               | 19    | 22     |      |
|                          |                                                     | 6.0           | 13               | 16    | 19     |      |
| $t_{TLH}$ ,<br>$t_{THL}$ | Maximum Output Transition Time, Any Output          | 2.0           | 75               | 95    | 110    | ns   |
|                          |                                                     | 3.0           | 27               | 32    | 36     |      |
|                          |                                                     | 4.5           | 15               | 19    | 22     |      |
|                          |                                                     | 6.0           | 13               | 16    | 19     |      |
| $C_{in}$                 | Maximum Input Capacitance                           |               | 10               | 10    | 10     | pF   |

|          |                                            |                                                                   |    |
|----------|--------------------------------------------|-------------------------------------------------------------------|----|
| $C_{PD}$ | Power Dissipation Capacitance (Per Buffer) | Typical @ 25°C, $V_{CC} = 5.0 \text{ V}$ , $V_{EE} = 0 \text{ V}$ | pF |
|          |                                            | 22                                                                |    |

Note 1:  $C_{PD}$  is defined as the value of the internal equivalent capacitance which is calculated from the operating current consumption without load. Average operating current can be obtained by the equation.

$$I_{CC(opr)} = C_{PD} \times V_{CC} \times f_{IN} + I_{CC}/6 \text{ (per gate)}$$

(number of gates is 4 for 74HC00A)

